MTH211: Theory of Statistics

Lecture 31: SLR model - Least Squares Estimation of parameters.

True Regression model. $E(Y|\underset{\sim}{x}) = \beta_0 + \beta_1 x$.

Y: Response variable (Regressand)

X: Explanatory variable (Regressor/ Predictor)

Data on n pairs of observations (y_i, x_i), $i=1(1)n$.

Random responses for different given values of the predictor

* $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i=1(1)n$. (i^{th} realization)

Random errors ϵ_i's are such that $E(\epsilon_i|x_i) = 0$

$$Cov(\epsilon_i, \epsilon_{i'}|x_i, x_{i'}) = \begin{cases} \sigma^2 & \text{if } i=i' \\ 0 & \text{if } i \neq i' \end{cases} \qquad (**)$$

* This is a simple (only 1 regressor), linear (in parameters) regression (SLR) model.

(**) $\Rightarrow$ $\underbrace{\text{Homoscedastic \& uncorrelated error}}_{\text{may be relaxed}}$ with $\underbrace{\text{zero mean}}_{\text{always in place}}$ WHY?

Least squares or L_2 norm minimization estimation method

Minimize the Sum of Squares of Error $= \sum_i \epsilon_i^2 = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ wrt β_0 & β_1

$= SSE(\beta_0, \beta_1)$, say.

Min. $SSE(\beta_0, \beta_1)$ is achieved at $\hat\beta_0$ and $\hat\beta_1$, the least sq. estimates of β_0 & β_1.

ie, $\left.\frac{\partial SSE(\beta_0,\beta_1)}{\partial \beta_0}\right|_{\hat\beta_0,\hat\beta_1} = 0$ $\Rightarrow$ $\left.\frac{\partial}{\partial \beta_0} \sum (y_i - \beta_0 - \beta_1 x_i)^2\right|_{\hat\beta_0,\hat\beta_1} = 0 \rightarrow (i)$

$\left.\frac{\partial SSE(\beta_0,\beta_1)}{\partial \beta_1}\right|_{\hat\beta_0,\hat\beta_1} = 0$ $\quad$ $\left.\frac{\partial}{\partial \beta_1} \sum (y_i - \beta_0 - \beta_1 x_i)^2\right|_{\hat\beta_0,\hat\beta_1} = 0 \rightarrow (ii)$

From (i), $\left.\begin{array}{l} n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i \\ \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum y_i x_i \end{array}\right\}$ 2 normal equations in 2 unknowns.

Solving, $\hat\beta_1 = \frac{\sum (y_i - \bar y)(x_i - \bar x)}{\sum (x_i - \bar x)^2} = \frac{s_{xy}}{s_{xx}}$ and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$, (CHECK this)

these are the least squares estimates of β_0 & β_1 from the given data.

On the other hand, $\hat\beta_0 = \bar Y - \hat\beta_1 \bar x$ & $\hat\beta_1 = \frac{\sum (Y_i - \bar Y)(x_i - \bar x)}{\sum (x_i - \bar x)^2}$ are the least squares estimators, these are statistics - fns of random variables $Y_1, \ldots, Y_n$, hence random variables themselves.

Important: We need to check, the above method actually minimizes (and not maximizes) the sum of squares error!

So, check whether the Hessian matrix $\left(\left(\frac{\partial^2 SSE(\beta_0,\beta_1)}{\partial\beta_0\,\partial\beta_1}\right)\right)_{\hat\beta_0,\hat\beta_1}$ is pd or not

Properties of the least squares estimators (the statistics) $\hat\beta_0$ & $\hat\beta_1$.

$$E(\hat\beta_1) = \frac{E\,\Sigma(y_i-\bar y)(x_i-\bar x)}{\Sigma(x_i-\bar x)^2} = \beta_1$$

$$E(\hat\beta_0) = E(\bar y) - E(\hat\beta_1)\bar x = \beta_0$$

check.

$\Rightarrow \hat\beta_0$ & $\hat\beta_1$ are unbiased estimators of β_0 & β_1, respectively

Further,
$$Var(\hat\beta_1) = Var\left(\frac{S_{xy}}{S_{xx}}\right) = \frac{\sigma^2}{S_{xx}}$$

$$Var(\hat\beta_0) = Var(\bar y) + \bar x^2 Var(\hat\beta_1) - 2\bar x\, Cov(\bar y,\hat\beta_1) = \sigma^2\left(\frac{1}{n}+\frac{\bar x^2}{S_{xx}}\right)$$

$$Cov(\hat\beta_0,\hat\beta_1) = -\frac{\bar x}{S_{xx}}\sigma^2$$

Check

Let the fitted values of y_i be denoted by $\hat y_i = \hat\beta_0 + \hat\beta_1 x_i$.
Difference between the observed (y_i) & fitted $(\hat y_i)$ responses gives an estimate of the error, called the residual.

So, the ith residual is $\hat\epsilon_i = y_i - \hat y_i$, $i=1(1)n$

Like y_i, $\hat y_i$ & $\hat\epsilon_i$ are also random variables. (with observed values from the data & the model fit)

Residuals (from the fit) play an important role in the estimation of the unknown error variance, σ^2.

Sum of squares Residuals $= \Sigma\hat\epsilon_i^2$, can be splitted up into Sum of Squares Total (corrected) and Sum of Squares Regression (or Model), where

$$SSTotal\,(corrected) = \sum_{i=1}^{n}(y_i-\bar y)^2, \quad SSRes = \sum_i (y_i-\hat y_i)^2, \text{ giving}$$

$$SSReg = \Sigma(\hat y_i - \bar y)^2, \quad \text{(check)}.$$

Then, it can be verified that $E\left(\Sigma(y_i-\hat y_i)^2\right) = (n-2)\sigma^2$, giving an unbiased estimator of σ^2, viz.,

$$\hat\sigma^2 = \frac{SS\,Res}{n-2} = \text{Mean Sq. Residuals}.$$